

Assignment #2 - Wireshark

- **Wireshark Lab - Getting Started**

- I used intro-wireshark-trace1.pcap from the wireshark-traces-9e.zip folder to answer the following 6 questions
- **Question 1:** Which of the following protocols are shown as appearing (i.e., are listed in the Wireshark "protocol" column) in your trace file: TCP, QUIC, HTTP, DNS, UDP, TLSv1.2?
 - **Answer:** HTTP, TCP, and TLSv1.2 are present. ARP, ICMPv6, and STP are also shown but are not listed in this question.

278	8.295669	Maxlinear_80:00:00	Apple_98:d9:27	ARP
279	8.295782	Apple_98:d9:27	Maxlinear_80:00:00	ARP
286	8.472728	10.0.0.44	128.119.245.12	HTTP
288	8.501613	128.119.245.12	10.0.0.44	HTTP
290	8.599454	10.0.0.44	128.119.245.12	HTTP
292	8.623422	128.119.245.12	10.0.0.44	HTTP
2	0.307157	fe80::a4f1:94ff:fec...	ff02::1	ICMPv6
239	3.276735	fe80::a4f1:94ff:fec...	ff02::1	ICMPv6
275	6.348795	fe80::a4f1:94ff:fec...	ff02::1	ICMPv6
295	9.318585	fe80::a4f1:94ff:fec...	ff02::1	ICMPv6
651	12.390344	fe80::a4f1:94ff:fec...	ff02::1	ICMPv6
1	0.000000	Sonos_25:3a:2a	Nearest-Customer-Br...	STP
3	1.023952	Sonos_25:3a:2a	Nearest-Customer-Br...	STP
4	2.047898	Sonos_25:3a:2a	Nearest-Customer-Br...	STP
5	3.072029	Sonos_25:3a:2a	Nearest-Customer-Br...	STP
240	3.993686	Sonos_25:3a:2a	Nearest-Customer-Br...	STP
243	5.017529	Sonos_25:3a:2a	Nearest-Customer-Br...	STP

641	12.241601	10.0.0.44	23.38.112.64	TCP
642	12.241786	10.0.0.44	23.38.112.64	TCP
643	12.242074	10.0.0.44	23.38.112.64	TCP
644	12.242215	10.0.0.44	23.38.112.64	TCP
645	12.242395	10.0.0.44	23.38.112.64	TCP
646	12.242529	10.0.0.44	23.38.112.64	TCP
647	12.243124	10.0.0.44	23.38.112.64	TCP
648	12.243271	10.0.0.44	23.38.112.64	TCP
649	12.243568	10.0.0.44	23.38.112.64	TCP
650	12.243765	10.0.0.44	23.38.112.64	TCP
6	3.182652	10.0.0.44	23.38.112.64	TLSv1.2
24	3.204758	23.38.112.64	10.0.0.44	TLSv1.2
26	3.205448	23.38.112.64	10.0.0.44	TLSv1.2
27	3.205453	23.38.112.64	10.0.0.44	TLSv1.2
28	3.205454	23.38.112.64	10.0.0.44	TLSv1.2
29	3.205458	23.38.112.64	10.0.0.44	TLSv1.2
30	3.205459	23.38.112.64	10.0.0.44	TLSv1.2

- **Question 2:** How long did it take from when the HTTP GET message was sent until the HTTP OK replay was received? (By default, the value of the Time column in the packet-listing window is the amount of time, in seconds, since Wireshark tracing began). To display the Time field in time-of-day format, select the Wireshark View pull down menu, then select Time Display Format, then select Time-of-day

- **Answer:** It took 28885 microseconds (approximately 0.03 seconds or about 28.9 milliseconds)

286	14:22:53.911521	10.0.0.44	128.119.245.12	HTTP	904	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
288	14:22:53.940406	128.119.245.12	10.0.0.44	HTTP	504	HTTP/1.1 200 OK (text/html)

- **Question 3:** What is the Internet address of the [gaia.umass.edu](http://www-net.cs.umass.edu) (www-net.cs.umass.edu)? What is the Internet address of your computer (if you are using the trace file) or the computer that sent the HTTP GET message?

- **Answer:** IP of computer → 10.0.0.44; IP of [gaia.umass.edu](http://www-net.cs.umass.edu) → 128.119.245.12

- Note: The computer IP is not the public IP, but rather the private IP

286	14:22:53.911521	10.0.0.44	128.119.245.12	HTTP	904	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
-----	-----------------	-----------	----------------	------	-----	---

- **Question 4:** Expand the information on the HTTP message in the Wireshark "Details of selected packet" window so you can see the fields in the HTTP GET request message. What type of Web browser issued the HTTP request? The answer is shown at the right end of the information following the "User-Agent:" field in the expanded HTTP message display. This field value in the HTTP message is how a web server learns what type of browser you are using. E.g. Firefox, Safari, Microsoft Internet Edge, Other

- **Answer:** Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:84.0) Gecko/20100101 Firefox/84.0

```

v Hypertext Transfer Protocol
  > GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:84.0) Gecko/20100101 Firefox/84.0\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\n
    Accept-Language: en-US,en;q=0.5\r\n
    Accept-Encoding: gzip, deflate\r\n
    DNT: 1\r\n
    Connection: keep-alive\r\n
  > Cookie [...]: _ga=GA1.2.539094814.1610028235; _fbp=fb.1.1581132068304.462218827; _hjid=721b807f-ada0
    Upgrade-Insecure-Requests: 1\r\n
    \r\n
    [Response in frame: 288]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html]

```

- **Question 5:** Expand the information on the Transmission Control Packet for this packet in the Wireshark "Details of selected packet" window so you can see the fields in the TCP segment carrying the HTTP message. What is the destination port number (the number following "Dest Port:" for the TCP segment containing the HTTP request) to which this HTTP request is being sent??

- **Answer:** This is being sent to port 80 (the default for HTTP).

```

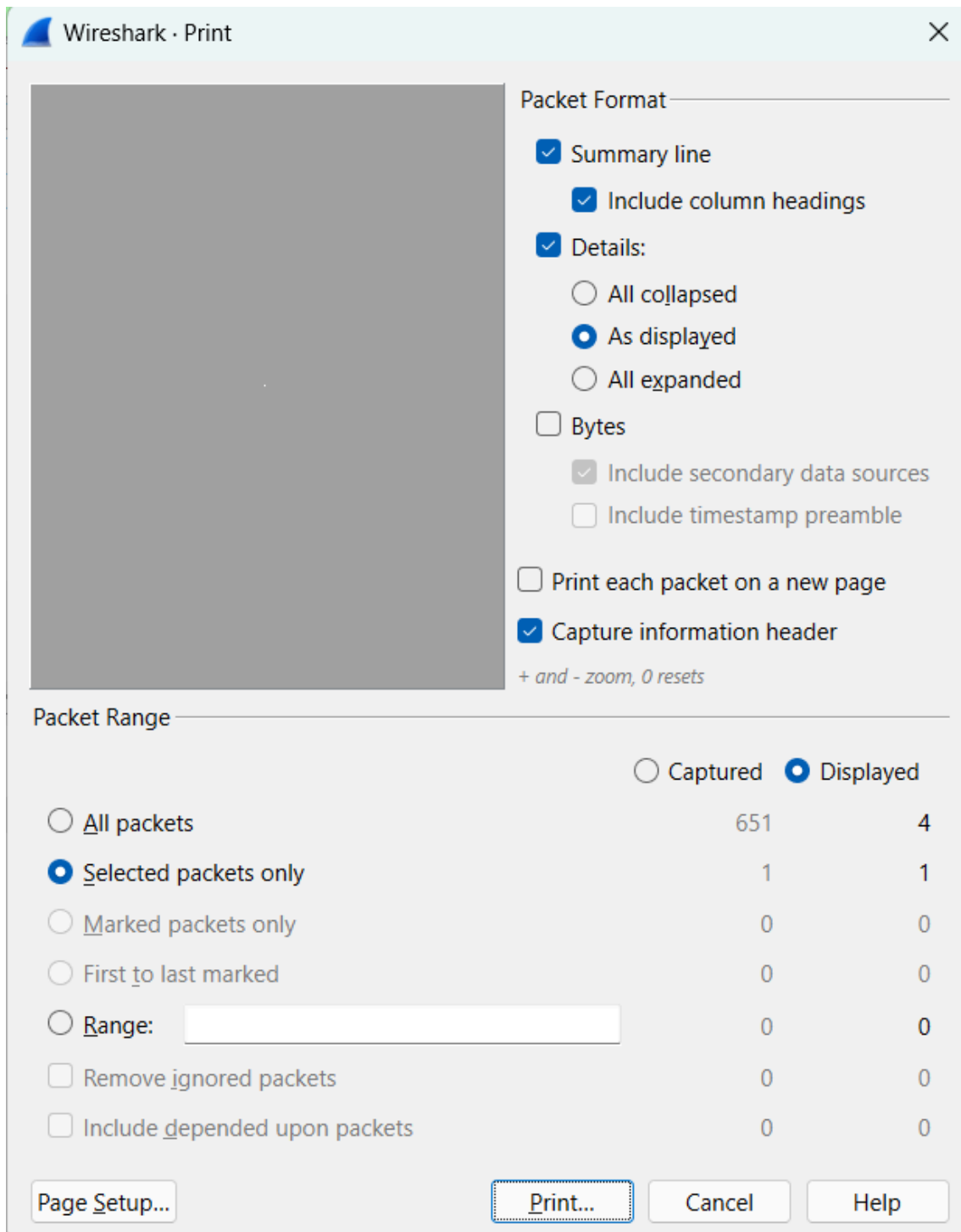
v Transmission Control Protocol, Src Port: 53962, Dst Port: 80, Seq: 1, Ack: 1, Len: 838
  Source Port: 53962
  Destination Port: 80
  [Stream index: 5]
  [Stream Packet Number: 4]

```

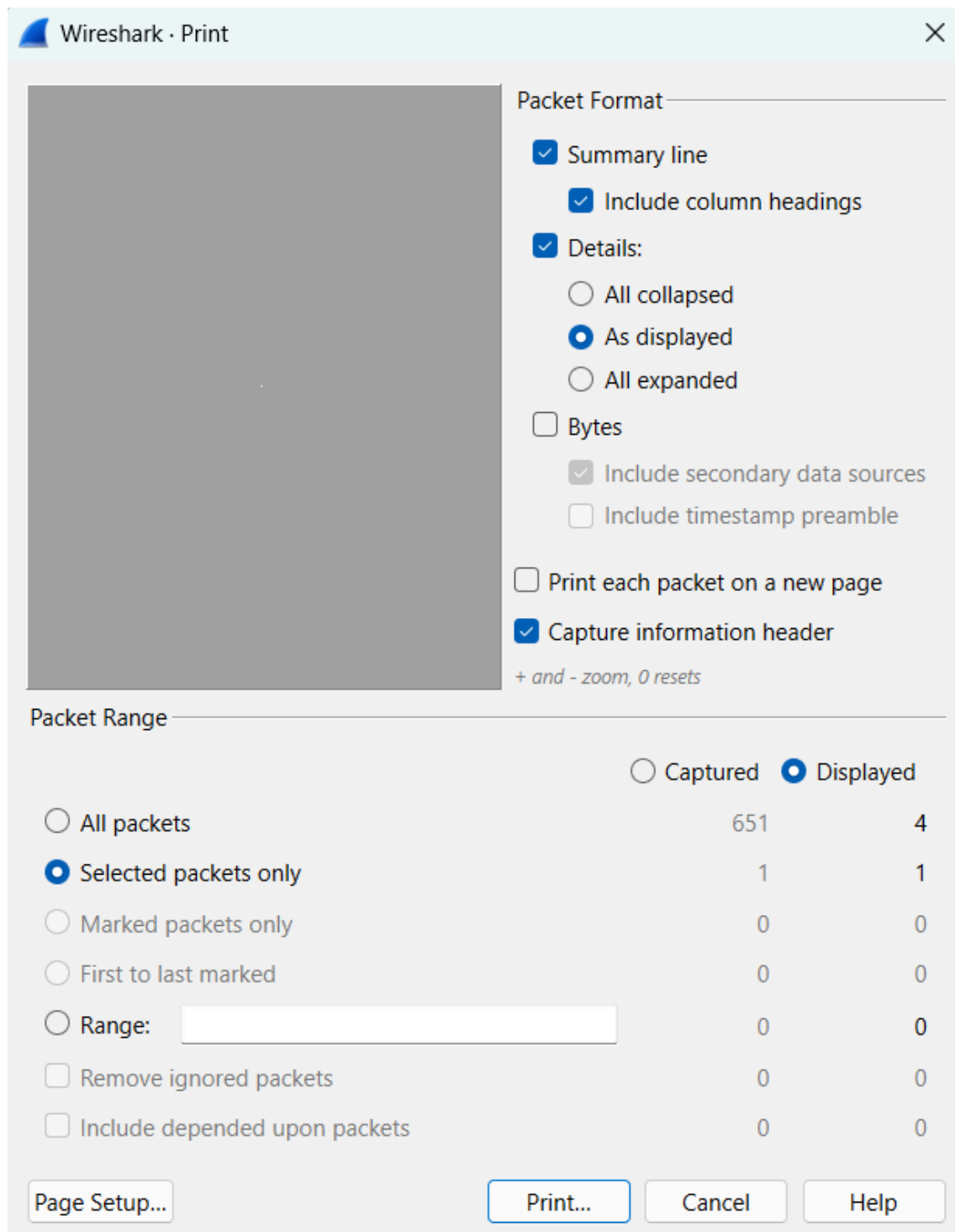
- **Question 6:** Print the two HTTP messages (GET and OK) referred to in Question 2. To do so, select Print from the Wireshark File command menu,

and select the "Selected Packet Only" and "Print as displayed" radial buttons, and then click OK.

▪ **Answer:**



GET Packet



OK Packet

- **Wireshark Lab: HTTP**

- I used the http-wireshark-trace1-1 trace file in the wireshark-traces-9e.zip folder to answer the following 7 questions

- **Question 1:** Is your browser running HTTP version 1.0, 1.1, or 2? What version of HTTP is the server running?

- **Answer:** My browser is using HTTP/1.1, located in the Hypertext Transfer Protocol section of the packet at the HTTP request method header.

```
▼ Hypertext Transfer Protocol
  > GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n
```

- The server is running HTTP/1.1 as well, also located in the Hypertext Transfer Protocol section of the packet at the HTTP response method header

```
▼ Hypertext Transfer Protocol
  > HTTP/1.1 200 OK\r\n
```

- **Question 2:** What languages (if any) does your browser indicate that it can accept to the server?

- **Answer:** Our browser can accept English (and prefers it), but also General English. This is located in the Hypertext Transfer Protocol layer section of the packet at the “Accept-Language” header.

```
▼ Hypertext Transfer Protocol
  > GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Cache-Control: max-age=0\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_4) AppleWebKit/537.36 (KHTML, like Gecko) C
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    \r\n
```

- **Question 3:** What is the IP address of your computer? What is the IP address of the gaia.cs.umass.edu server?

- **Answer:** The IP address of my computer is 10.0.0.44 and the IP address of the gaia.cs.umass.edu server is 128.119.245.12. These were located in the IPv4 section of the packet, underneath the Source and Destination Address headers.

```

  ▾ Internet Protocol Version 4, Src: 10.0.0.44, Dst: 128.119.245.12
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      Total Length: 559
      Identification: 0x0000 (0)
    > 010. .... = Flags: 0x2, Don't fragment
      ...0 0000 0000 0000 = Fragment Offset: 0
      Time to Live: 64
      Protocol: TCP (6)
      Header Checksum: 0xb919 [validation disabled]
      [Header checksum status: Unverified]
      Source Address: 10.0.0.44
      Destination Address: 128.119.245.12
      [Stream index: 0]

```

- **Question 4:** What is the status code returned from the server to your browser?

- **Answer:** The status code returned from the server to the browser is HTTP/1.1 **200** OK. This is located in the Hypertext Transfer Protocol section of the packet, in the HTTP response method header.

```

  ▾ Hypertext Transfer Protocol
    > HTTP/1.1 200 OK\r\n

```

- **Question 5:** When was the HTML file that you are retrieving last modified at the server?

- **Answer:** The HTML file that we are retrieving from the server was last modified on Saturday, January 30th, 2021 at 06:59:02 GMT. This is located in the Hypertext Transfer Protocol section of the packet underneath the "Last-Modified" header.

```

  ▾ Hypertext Transfer Protocol
    > HTTP/1.1 200 OK\r\n
      Date: Sat, 30 Jan 2021 21:43:30 GMT\r\n
      Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.14 mod_perl/2.0.11 Perl/v5.16.3\r\n
      Last-Modified: Sat, 30 Jan 2021 06:59:02 GMT\r\n
      ETag: "80-5ba18a7e1c636"\r\n

```

- **Question 6:** How many bytes of content are being returned to your browser?

- **Answer:** 128 bytes of content are being returned to my browser. This can be found in the Hypertext Transfer Protocol section of the packet,

underneath the "Content-Length" header.

```
> HTTP/1.1 200 OK\r\n
Date: Sat, 30 Jan 2021 21:43:30 GMT\r\n
Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.14 mod_perl/2.0.11 Perl/v5.16.3\r\n
Last-Modified: Sat, 30 Jan 2021 06:59:02 GMT\r\n
ETag: "80-5ba18a7e1c636"\r\n
Accept-Ranges: bytes\r\n
✓ Content-Length: 128\r\n
  [Content length: 128]
Keep-Alive: timeout=5, max=100\r\n
Connection: Keep-Alive\r\n
Content-Type: text/html; charset=UTF-8\r\n
\r\n
```

- **Question 7:** By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name one.
 - **Answer:** No, there are no extra headers that are not displayed in the packet window. The packet listing (the single line) only displays high-level information, but everything in the packet listing is in the packet content window.
- **I used the http-wireshark-trace2-1 trace file in the wireshark-traces-9e.zip folder to answer the following 4 questions**
- **Question 8:** Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?
 - **Answer:** No, there is no "IF-MODIFIED-SINCE" line in the first HTTP GET.

```
> Frame 56: Packet, 547 bytes on wire (4376 bits), 547 bytes captured (4376 bits) on interface en0, id 0
> Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Maxlinear_80:00:00 (00:50:f1:80:00:00)
> Internet Protocol Version 4, Src: 10.0.0.44, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 52895, Dst Port: 80, Seq: 1, Ack: 1, Len: 481
✓ Hypertext Transfer Protocol
  > GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_4) AppleWebKit/537.36 (KHTML, like Gecko) C
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    \r\n
    [Response in frame: 58]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]
```

- **Question 9:** Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?
 - **Answer:** Yes, the server explicitly returns the contents of the file. We can tell by the HTTP/1.1 200 OK response code returning an HTML file under the "Line-based text data: text/html" part of the packet.

```

Content-Type: text/html; charset=UTF-8\r\n
\r\n
[Request in frame: 56]
[Time since request: 24.609000 milliseconds]
[Request URI: /wireshark-labs/HTTP-wireshark-file2.html]
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]
File Data: 371 bytes
▼ Line-based text data: text/html (10 lines)
\r\n
<html>\r\n
\r\n
Congratulations again! Now you've downloaded the file lab2-2.html. <br>\n
This file's last modification date will not change. <p>\n
Thus if you download this multiple times on your browser, a complete copy <br>\n
will only be sent once by the server due to the inclusion of the IN-MODIFIED-SINCE<br>\n
field in your browser's HTTP GET request to the server.\n
\r\n
</html>\n

```

- **Question 10:** Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE" header?
 - **Answer:** Yes, there is an "IF-MODIFIED-SINCE" line the in second HTTP GET. The information following this header is the date time that we are checking if the file was modified since. The date time provided is "Sat, 30 Jan 2021 06:59:02 GMT", underneath the "IF-MODIFIED-SINCE" header in the Hypertext Transfer Protocol section of the packet.

```

> Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Maxlinear_80:00:00 (00:50:f1:80:00:00)
> Internet Protocol Version 4, Src: 10.0.0.44, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 52895, Dst Port: 80, Seq: 482, Ack: 731, Len: 593
v Hypertext Transfer Protocol
  > GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Cache-Control: max-age=0\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_4) AppleWebKit/537.36 (KHTML, like Gecko)
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    If-None-Match: "173-5ba18a7e1ba7e"\r\n
    If-Modified-Since: Sat, 30 Jan 2021 06:59:02 GMT\r\n
    \r\n
    [Response in frame: 556]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]

```

- o **Question 11:** What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.
 - **Answer:** The HTTP status code and phrase returned from the server in response to the second HTTP GET is HTTP/1.1 304 Not Modified. The server does not explicitly return the contents of the file and we can tell because there is no message body. This happens because the browser has cached the results from the first HTTP/1.1 200 OK response, when it explicitly sent the contents of the file. Since the file was not modified since then, the browser can safely assume that the cached data contains current information.

```

v Hypertext Transfer Protocol
  v HTTP/1.1 304 Not Modified\r\n
    Response Version: HTTP/1.1
    Status Code: 304
    [Status Code Description: Not Modified]
    Response Phrase: Not Modified
    Date: Sat, 30 Jan 2021 18:19:56 GMT\r\n
    Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.14 mod_perl/2.0.11 Perl/v5.16.3\r\n
    Connection: Keep-Alive\r\n
    Keep-Alive: timeout=5, max=99\r\n
    ETag: "173-5ba18a7e1ba7e"\r\n
    \r\n
    [Request in frame: 555]
    [Time since request: 22.630000 milliseconds]
    [Request URI: /wireshark-labs/HTTP-wireshark-file2.html]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]

```

- o I used the http-wireshark-trace3-1 trace file in the wireshark-traces-9e.zip folder to answer the following 4 questions

- **Question 12:** How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill of Rights?

- **Answer:** The browser sent 1 HTTP GET message. Packet #26 contains the GET message for the Bill of Rights.

```
> Frame 26: Packet, 547 bytes on wire (4376 bits), 547 bytes captured (4376 bits) on interface en0, id 1
> Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Maxlinear_80:00:00 (00:50:f1:80:00:00)
> Internet Protocol Version 4, Src: 10.0.0.44, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 54985, Dst Port: 80, Seq: 1, Ack: 1, Len: 481
  Hypertext Transfer Protocol
    GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1\r\n
      Request Method: GET
      Request URI: /wireshark-labs/HTTP-wireshark-file3.html
      Request Version: HTTP/1.1
      Host: gaia.cs.umass.edu\r\n
      Connection: keep-alive\r\n
      Upgrade-Insecure-Requests: 1\r\n
      User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_4) AppleWebKit/537.36 (KHTML, like Gecko)
      Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*
      Accept-Encoding: gzip, deflate\r\n
      Accept-Language: en-US,en;q=0.9\r\n
      \r\n
```

- **Question 13:** Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

- **Answer:** Packet #32 contains the status code and phrase associated with the response to the HTTP GET request.

```
> Frame 32: Packet, 583 bytes on wire (4664 bits), 583 bytes captured (4664 bits) on interface en0, id 1
> Ethernet II, Src: Maxlinear_80:00:00 (00:50:f1:80:00:00), Dst: Apple_98:d9:27 (78:4f:43:98:d9:27)
> Internet Protocol Version 4, Src: 128.119.245.12, Dst: 10.0.0.44
> Transmission Control Protocol, Src Port: 80, Dst Port: 54985, Seq: 4345, Ack: 482, Len: 517
> [4 Reassembled TCP Segments (4861 bytes): #28(1448), #29(1448), #31(1448), #32(517)]
  Hypertext Transfer Protocol
    HTTP/1.1 200 OK\r\n
      Response Version: HTTP/1.1
      Status Code: 200
      [Status Code Description: OK]
      Response Phrase: OK
      Date: Sat, 30 Jan 2021 21:45:42 GMT\r\n
      Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.14 mod_perl/2.0.11 Perl/v5.16.3\r\n
      Last-Modified: Sat, 30 Jan 2021 06:59:02 GMT\r\n
      ETag: "1194-5ba18a7e16875"\r\n
      Accept-Ranges: bytes\r\n
      Content-Length: 4590\r\n
```

- **Question 14:** What is the status code and phrase in the response?

- **Answer:** The status code and phrase in the response is HTTP/1.1 200 OK.

```

> Frame 32: Packet, 583 bytes on wire (4664 bits), 583 bytes captured
> Ethernet II, Src: Maxlinear_80:00:00 (00:50:f1:80:00:00), Dst: 
> Internet Protocol Version 4, Src: 128.119.245.12, Dst: 10.0.0.4
> Transmission Control Protocol, Src Port: 80, Dst Port: 54985, S
> [4 Reassembled TCP Segments (4861 bytes): #28(1448), #29(1448),
  ▾ Hypertext Transfer Protocol
    ▾ HTTP/1.1 200 OK\r\n
      Response Version: HTTP/1.1
      Status Code: 200
      [Status Code Description: OK]
      Response Phrase: OK

```

- **Question 15:** How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?
 - **Answer: Four** data-containing TCP (only) segments were needed to carry the single HTTP response and the text of the Bill of Rights. If the question means how many packets of TCP being the highest protocol layer used were needed to transfer the data, it would be three. We can tell by looking at the Wireshark Packet Listings, where each one says [TCP PDU reassembled in 32]. However, if we add up the length of each $1448 + 1448 + 1448 = 4344$ and compare it to the byte count of the file data (4500), we see that we are missing 156 bytes. These 156 bytes come from Packet #32 itself, inside the Transmission Control Protocol section of the packet. We can find this as well in a section titled [4 Reassembled TCP Segments (4861 bytes): #28(1448), #29(1448), #31(1448), and #32(517)]

```

  ▾ [4 Reassembled TCP Segments (4861 bytes): #28(1448), #29(1448),
    [Frame: 28, payload: 0-1447 (1448 bytes)]
    [Frame: 29, payload: 1448-2895 (1448 bytes)]
    [Frame: 31, payload: 2896-4343 (1448 bytes)]
    [Frame: 32, payload: 4344-4860 (517 bytes)]
    [Segment count: 4]
    [Reassembled TCP length: 4861]
    [Reassembled TCP Data [...]: 485454502f312e3120323030204f4b0d0a

```

```

  [TCP PDU reassembled in 32]
584 [TCP PDU reassembled in 32]
0
584 [TCP PDU reassembled in 32]

```

- I used the **http-wireshark-trace4-1** trace file in the **wireshark-traces-9e.zip** folder to answer the following 2 questions

- **Question 16:** How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

- **Answer:** Our browser sent **four** GET requests. We sent the initial GET to the "wireshark-file4.html", a second GET to retrieve /pearson.png, and then a third GET to get /8E_cover_small. I'm unsure if this was intended, but a fourth GET request was sent to the "https" version of the /8E_cover_small.jpg URI as the third GET was responded to with a HTTP/1.1 301 Moved Permanently, which redirected to a "https" version. This fourth GET request was a base64 encoded OSCP request.

95	16:53:03.808469	10.0.0.44	128.119.245.12	HTTP	547	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
97	16:53:03.838833	128.119.245.12	10.0.0.44	HTTP	1367	HTTP/1.1 200 OK (text/html)
99	16:53:03.862605	10.0.0.44	128.119.245.12	HTTP	493	GET /pearson.png HTTP/1.1
104	16:53:03.883040	128.119.245.12	10.0.0.44	HTTP	781	HTTP/1.1 200 OK (PNG)
118	16:53:04.100178	10.0.0.44	178.79.137.164	HTTP	500	GET /8E_cover_small.jpg HTTP/1.1
120	16:53:04.242092	178.79.137.164	10.0.0.44	HTTP	237	HTTP/1.1 301 Moved Permanently
144	16:53:04.547953	10.0.0.44	104.98.115.146	HTTP	361	GET /MFgwVqADAgEAME8wTTBLMAkGBSs0AwIaBQAEFEjayaD7K9MtT%2
155	16:53:04.634747	104.98.115.146	10.0.0.44	OCSP	955	Response

- The Internet addresses these GET requests were sent to are:
 - gaia.cs.mass.edu → 128.119.245.12
 - /pearson.png → 128.119.245.12
 - /8E_cover_small.jpg → 178.79.137.164
 - Redirection to "https" → 104.98.115.146
- **Question 17:** Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain

- **Answer:** The two images were downloaded serially. We can tell by looking at the date/times that the GETs and responses were. The GET request for /pearson.png was sent at 16:53:03.862605 and the response to this GET was received at 16:53:03.883040. The GET request for /8E_cover_small.jpg was sent at 16:53:04.100178 and the response was received at 16:53:04.242092.

- I used the http-wireshark-trace5-1 trace file in the wireshark-traces-9e.zip folder to answer the following 2 questions
- **Question 18:** What is the server's response (status and phrase) in response to the initial HTTP GET message from your browser?
 - **Answer:** The server's response to the initial HTTP GET message from my browser was HTTP/1.1 401 Unauthorized. This is located in the Hypertext Transfer Protocol section of the packet, in the HTTP response method header.

```
▼ Hypertext Transfer Protocol
  ▼ HTTP/1.1 401 Unauthorized\r\n
    Response Version: HTTP/1.1
    Status Code: 401
    [Status Code Description: Unauthorized]
    Response Phrase: Unauthorized
    Date: Sat, 30 Jan 2021 22:04:57 GMT\r\n
    Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.14
    WWW-Authenticate: Basic realm="wireshark-students only"\r\n
  > Content-Length: 381\r\n
    Keep-Alive: timeout=5, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=iso-8859-1\r\n
    \r\n
```

- **Question 19:** When your browser sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?
 - **Answer:** A couple new fields are included such as the Cache-Control and Authorization headers. This is found in the Hypertext Transfer Protocol section of the packet.

▼ Hypertext Transfer Protocol

- ▼ GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.htm
Request Method: GET
Request URI: /wireshark-labs/protected_pages/HTTP-wireshark-file5.htm
Request Version: HTTP/1.1
Host: gaia.cs.umass.edu\r\n
Connection: keep-alive\r\n
Cache-Control: max-age=0\r\n
- ▼ Authorization: Basic d2lyZXNoYXJrLXN0dWR1bnRzOm5ldHdvcm5=\r\n
Credentials: wireshark-students:network
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_4) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/80.0.3987.163 Safari/537.36\r\n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9\r\n
\r\n
[Response in frame: 482]
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/protected_pages/HTTP-wireshark-file5.htm]